

PROFESSIONAL SUMMARY

LLM serving control-plane engineer with five years of cloud-native AI/HPC experience and nearly three years in production serving. Owns Kubernetes-native platform work for heterogeneous accelerator fleets across model lifecycle, P/D topology and capacity governance, Mooncake/RoCE KV transfer, and KV-aware routing; also delivered multi-tenant batch inference, model gateways, and production observability. Primary languages: Go and Python, with Rust routing experience.

EXPERIENCE

Zhejiang Lab · High-Performance Computing Facility Research Center

2024 - Present

Senior Research Specialist | LLM Serving / Model-as-a-Service (MaaS) Platform

- Owned the Kubernetes-native serving control plane for heterogeneous accelerator fleets, standardizing model-service declarations, worker lifecycle, P/D topology, layered readiness, and failure isolation across SGLang and vLLM backends.
- Productionized capacity governance around profiled serving curves, converting live request shapes into independent prefill/decode targets with auditable readiness gates, stabilization windows, and cooldown controls.
- Delivered OpenAI-compatible batch inference and model gateways with durable sharding, multi-tenant admission, recovery, and end-to-end tracing for production data-distillation and batch-generation workloads.

Zhejiang Lab · Intelligent Supercomputing Research Center

2021 - 2023

Engineering Specialist | Cloud-Native AI / HPC Platform

- Customized Kubernetes device-plugin and resource-management paths for heterogeneous accelerator discovery, node resource reporting, container-runtime validation, device-level allocation, and fault isolation.
- Adapted PyTorch/models and distributed job environments to Linux distributions and heterogeneous accelerator stacks, establishing the platform foundation for production serving and scheduling.

SELECTED SYSTEMS

Heterogeneous Runtime & KV-Aware Routing

Worker Bootstrap · KV Event · Mooncake / RoCE

- Separated process, model, KV-event, Mooncake/RoCE data-channel, and traffic readiness; admitted workers only after model loading, P/D bootstrap, transfer setup, and Router registration all converged.
- Used the KV event plane for cache-location metadata and Mooncake Transfer Engine over RoCE for KV-tensor movement, keeping KV data transfer out of the Router.
- Routed on prefix overlap, worker load, accelerator compatibility, and topology; removed unavailable P/D pairs and fell back to healthy paths on stale events or transfer failures.

SELECTED SYSTEMS (CONTINUED)

Adaptive P/D Planner and Scaling Guardrails	Capacity Profile · ScaleIntent · Guardrails
- Derived request rate, input/output tokens, prefill compute/cached tokens, KV hit rate, and queue state from rolling windows; selected independent capacity profiles by model, accelerator, quantization, TP/DP, and engine version. - Computed prefill/decode demand and safe replica targets independently, using the bottleneck as the scaling intent; retained request shape, selected profile points, and blockers for audit, with pending gates, one-step scaling, stabilization windows, cooldown, and stale-intent checks.	
Multi-Tenant Batch Inference Platform	OpenAI Batch API · Scheduling · Recovery
- Kept task truth in the database while schedulers/workers claimed, sharded, executed, renewed, and finalized work; combined Kubernetes Lease, request-level isolation, and failure-tolerant finalization for multi-replica recovery without duplicate execution. - Covered streaming JSONL, tenant/model concurrency budgets, cancellation, expiry cleanup, and output/error writeback from the OpenAI Batch API contract through production monitoring while limiting offline impact on online serving.	
Inference Router and Multi-Tenant Admission	P/D Routing · Priority Admission · Tracing
- Implemented P0-P3 admission in Rust with global/tier semaphores, bounded queues, strict priority, and per-tenant round-robin; held permits through streaming-response completion to avoid releasing capacity early. - Integrated SGLang HTTP/gRPC P/D routing and propagated bootstrap/transfer context; moved KV data directly through Mooncake/RoCE while correlating request IDs, trace context, queue state, and KV-transfer telemetry.	

CORE TECHNOLOGIES

Inference Runtime & Routing	SGLang, vLLM, Mooncake Transfer Engine, RoCE/RDMA, KV event plane, KV-aware routing, P/D disaggregation
Scheduling & Control Plane	Go, Kubernetes, CRDs/controllers, capacity profiles, P/D autoscaling, reconciliation, readiness, leader election, Helm, ArgoCD
Gateway & Async Serving	Python, Rust, OpenAI-compatible APIs, Batch API, LiteLLM, Higress, Redis, MySQL, multi-tenant concurrency, failure recovery
Observability & Performance	Prometheus, request tracing, QPS/TTFT/TPOT, ISL/OSL, KV hit/transfer, throughput curves, distributed-systems debugging

EDUCATION

Arizona State University	May 2019 - Dec 2020
Master of Computer Science, GPA 3.97 / 4.00	
Arizona State University	Aug 2015 - May 2019
B.S. in Computer Science	

PUBLICATION & IP

Third author, Prompt2Length: A Lightweight Model for Predicting Output Length in LLM Inference , IEEE AIAHPC 2025; co-inventor on two granted invention patents and contributor to two authorized software copyrights.	
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